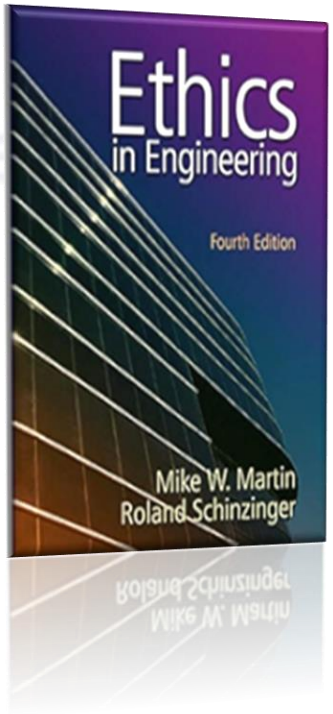
Engineering Ethics (HSS-422) (3+0)



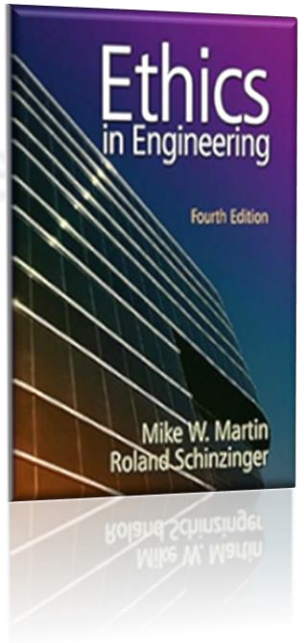




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CASE STUDY: PRIVACY BREACH AT HEALTHIFY SYSTEMS

Week No 15

By Engr. Syed Rizwan Ali

Learning Outcomes

In this Lecture will cover ..

- Understand the Importance of Ethical Decision-Making: Students will learn to recognize and prioritize ethical considerations in software development, understanding how ethical decision-making impacts the security and privacy of user data.
- Identify the Consequences of Ignoring Security Protocols: Students will be able to identify the risks and consequences associated with neglecting security measures, particularly in the development of applications that handle sensitive information.

Learning Outcomes

In this Lecture will cover ..

- Analyze the Role of Leadership in Ethical Practices: Students will analyze how leadership styles and company culture can influence the ethical behavior of teams, particularly in high-pressure situations involving tight deadlines and significant project demands.
- Evaluate Legal and Social Implications: Students will gain insights into the legal repercussions and social impacts of data breaches, learning about the responsibilities of companies towards their users and the potential societal consequences of negligence.

Learning Outcomes

In this Lecture will cover ..

- Develop Strategies for Ethical Conflict Resolution: Students will learn to devise strategies to handle ethical dilemmas, such as balancing transp

AN EXPLORATION OF ENGINEERING ETHICS IN SOFTWARE DEVELOPMENT



Case Study: Privacy Breach at Healthify Systems

- **Introduction:**

- Healthify Systems is a renowned software company known for its health analytics platforms used by hospitals and clinics worldwide. Recently, the company developed a new app, MedTrack, designed to monitor patient activities and provide predictive insights into patient health trends. MedTrack uses patient data such as location, health metrics, and personal identifiers.

Case Study: Privacy Breach at Healthify Systems

❖ **Scenario:**

- Three months after the launch, a whistleblower at Healthify Systems disclosed that the MedTrack app had severe vulnerabilities that could potentially expose patient data.

- The initial design and testing phase, stressed by tight deadlines and high expectations, led to significant oversight in data security and privacy protocols..

Case Study: Privacy Breach at Healthify Systems

❖ **Scenario:**

- Further investigation revealed that the development team had raised concerns about these vulnerabilities during the testing phase.

- However, due to pressure to meet launch deadlines and fear of reprisals from upper management, these concerns were not adequately addressed.

Case Study: Privacy Breach at Healthify Systems

❖ **Scenario:**

- The breach came to light when an independent security analyst discovered that patient data could be accessed through a basic security loophole.

- The news caused a public uproar, leading to legal consequences and a loss of trust in Healthify Systems.

Case Study: Privacy Breach at Healthify Systems

- **Ethical Dilemmas:**
- **Pressure vs. Integrity:** The development team faced pressure to meet deadlines which compromised their professional integrity.

- They chose to prioritize organizational demands over the ethical obligation to ensure software safety and privacy.

Case Study: Privacy Breach at Healthify Systems

❖ **Ethical Dilemmas:**

- **Transparency vs. Fear of Reprisal:** Employees were aware of the flaws but feared reprisal from management.

- This situation posed an ethical dilemma between transparency and self-preservation.

Case Study: Privacy Breach at Healthify Systems

❖ **Ethical Dilemmas:**

- **Responsibility for Reporting:** There was a collective responsibility among team members to report and address the vulnerabilities.

- The failure to do so raises questions about individual versus collective ethics in professional settings.

Case Study: Privacy Breach at Healthify Systems

❖ **Discussion Points:**

- **Role of Ethical Leadership:** How might leadership styles have influenced the team's decision to overlook the ethical concerns? Discuss the importance of ethical leadership in fostering a culture where employees feel safe to report issues.
- **Professional Responsibility:** Evaluate the professional responsibilities of software – continue.

Case Study: Privacy Breach at Healthify Systems

❖ **Discussion Points:**

- **Professional Responsibility:** Software engineers in terms of adhering to ethical standards and practices, particularly regarding user privacy and data protection.
- **Legal and Social Implications:** Explore the legal repercussions that could follow from such a breach. Discuss the social responsibility of companies in protecting user data and the potential societal impact of their failure to do so.

Case Study: Privacy Breach at Healthify Systems

❖ **Moral of the Case Study:**

- The moral of the case study on Healthify Systems and its privacy breach incident centers around the paramount importance of ethical responsibility and due diligence in software development.
- It underscores the necessity for organizations to prioritize the security and privacy of user data over business pressures such as meeting launch deadlines or achieving market firsts.

Case Study: Privacy Breach at Healthify Systems

❖ **Moral of the Case Study:**

- Additionally, it highlights the critical role of transparent and ethical leadership in fostering an environment where employees feel empowered and obligated to speak up about potential issues without fear of reprisal.
- This case serves as a reminder that the long-term trust and safety of users are integral to a company's success and reputation, far outweighing the short-term gains from cutting corners.

Case Study: Privacy Breach at Healthify Systems

❖ **Conclusion:**

- This case study serves as a crucial learning point for software engineers about the significance of ethical considerations in software development.
- It underscores the need for ethical vigilance, especially in fields where technology directly impacts human lives.

Thanks

Any Question

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